CS-229

Dynamics week!

- Classical dynamics
- SGD with momentum
- Adam

Review of classical dynamics

Momentum, oscillations, friction

Deterministic & stochastic

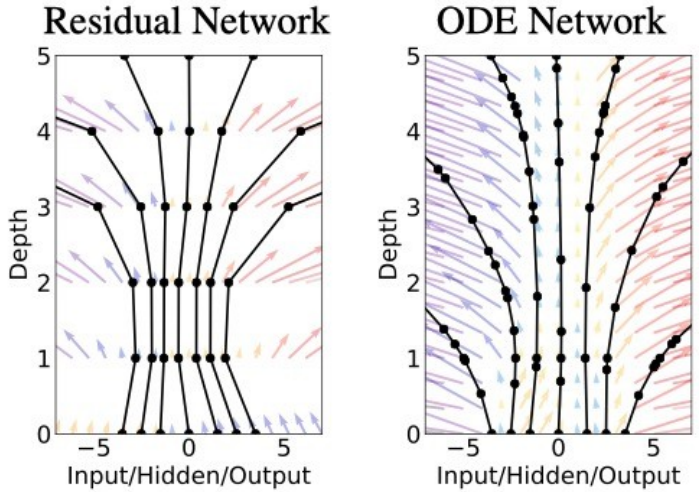
Hamiltonian formulation

And how it all relates to machine learning!

The high level takeaways without the technical stuff

- Gradient descent is bad, but *stochastic* gradient descent is good
  - Noise = exploration
  - More noise can come from larger learning rate or smaller batch size
  - Learning rate schedulers give good exploration at the beginning, stable convergence at the end
- Therefore, batch size and learning rate are (inversely) related
  - If you tune your code on a small machine, then run it on a bigger GPU and double the batch size, you should also double the learning rate.
- Momentum and Adaptive Momentum (Adam) are effective ways to accelerate learning

Mathematical themes



Continuous to discrete and vice versa (SGD, Neural ODE

Dynamic distributions (Flows, MCMC)

Dynamical systems concepts (fixed points, stationary distributions)

Classical physics (Momentum, oscillation, damping, Brownian motion, Hamiltonian dynamics)

Deterministic versus stochastic dynamics (GD vs SGD, Hamiltonian versus Langevin dynamics)

Gradient descent (!)

Consider a loss function, with weights, **w**:

$$L(\boldsymbol{w}) = \frac{1}{n}\sum_{i} l(\boldsymbol{w}, \boldsymbol{x}_i)$$

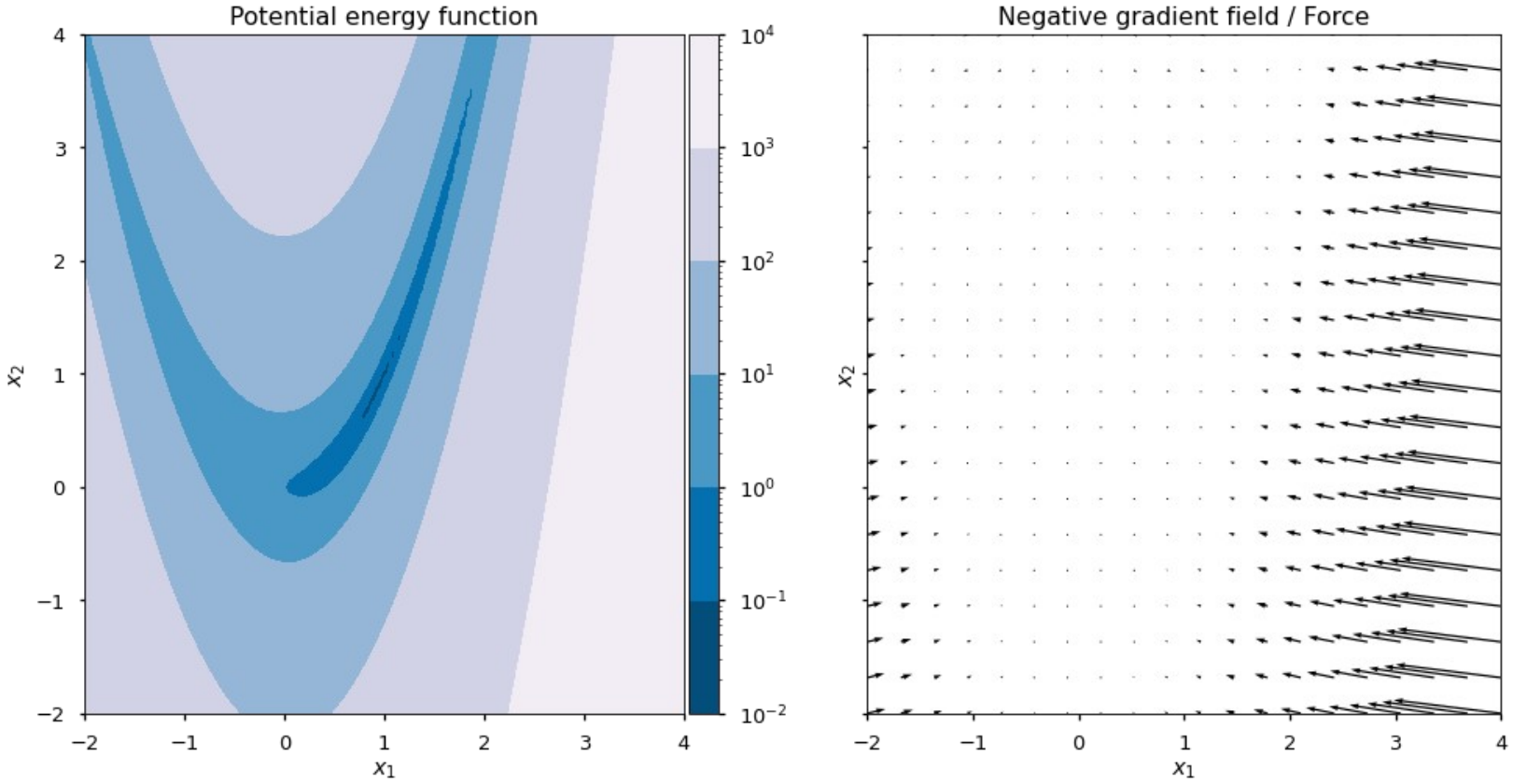
We use gradient descent to find a local minimum ("fixed point"):

$$\boldsymbol{w}_{t+\eta} = \boldsymbol{w}_t - \eta \boldsymbol{\nabla}_{\boldsymbol{w}} L(\boldsymbol{w}_t)$$

First recurring idea: going to the continuous limit!

$$\lim_{\eta \to 0} (\boldsymbol{w}_{t+\eta} - \boldsymbol{w}_t)/\eta = d\boldsymbol{w}/dt = \dot{\boldsymbol{w}} = -\boldsymbol{\nabla}_{\boldsymbol{w}} L(\boldsymbol{w}(t))$$

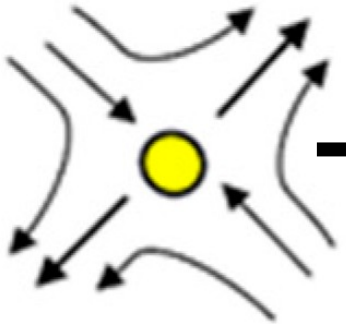
Example loss: Rosenbrock function



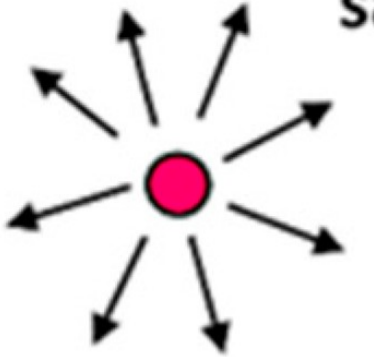
The local minimum is an “*attracting fixed point*” of the dynamics

•

Repeller / unstable fixed point

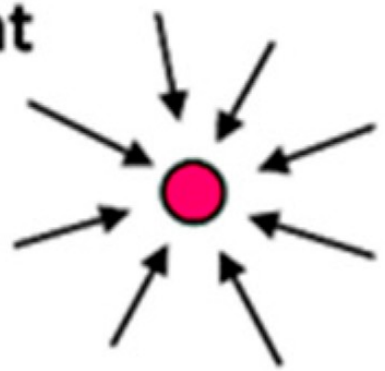


saddle point



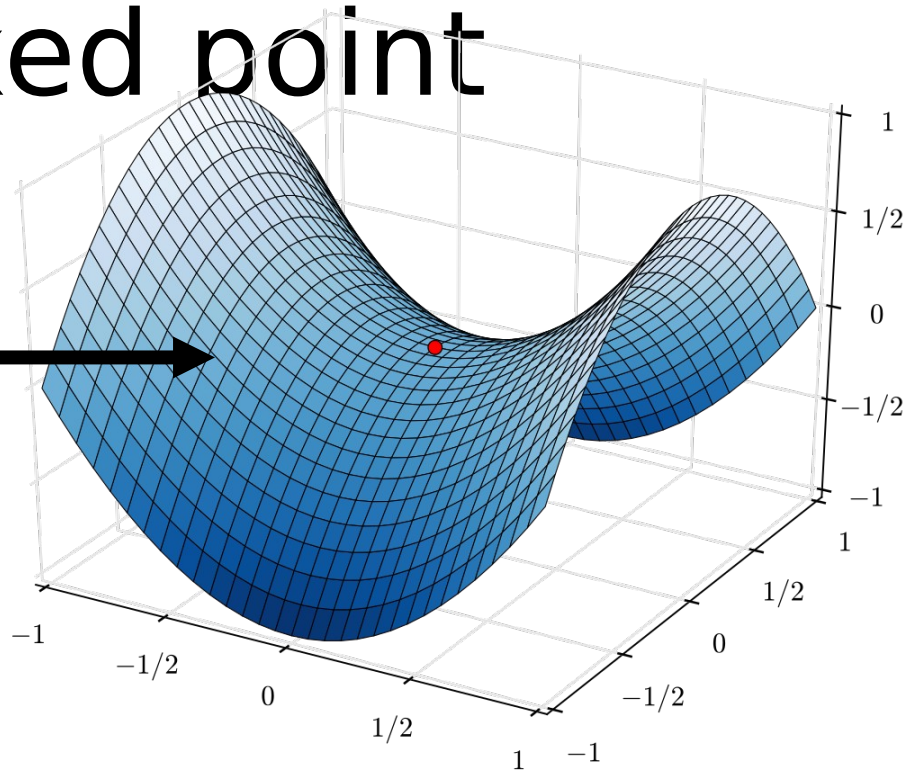
unstable node

repelling

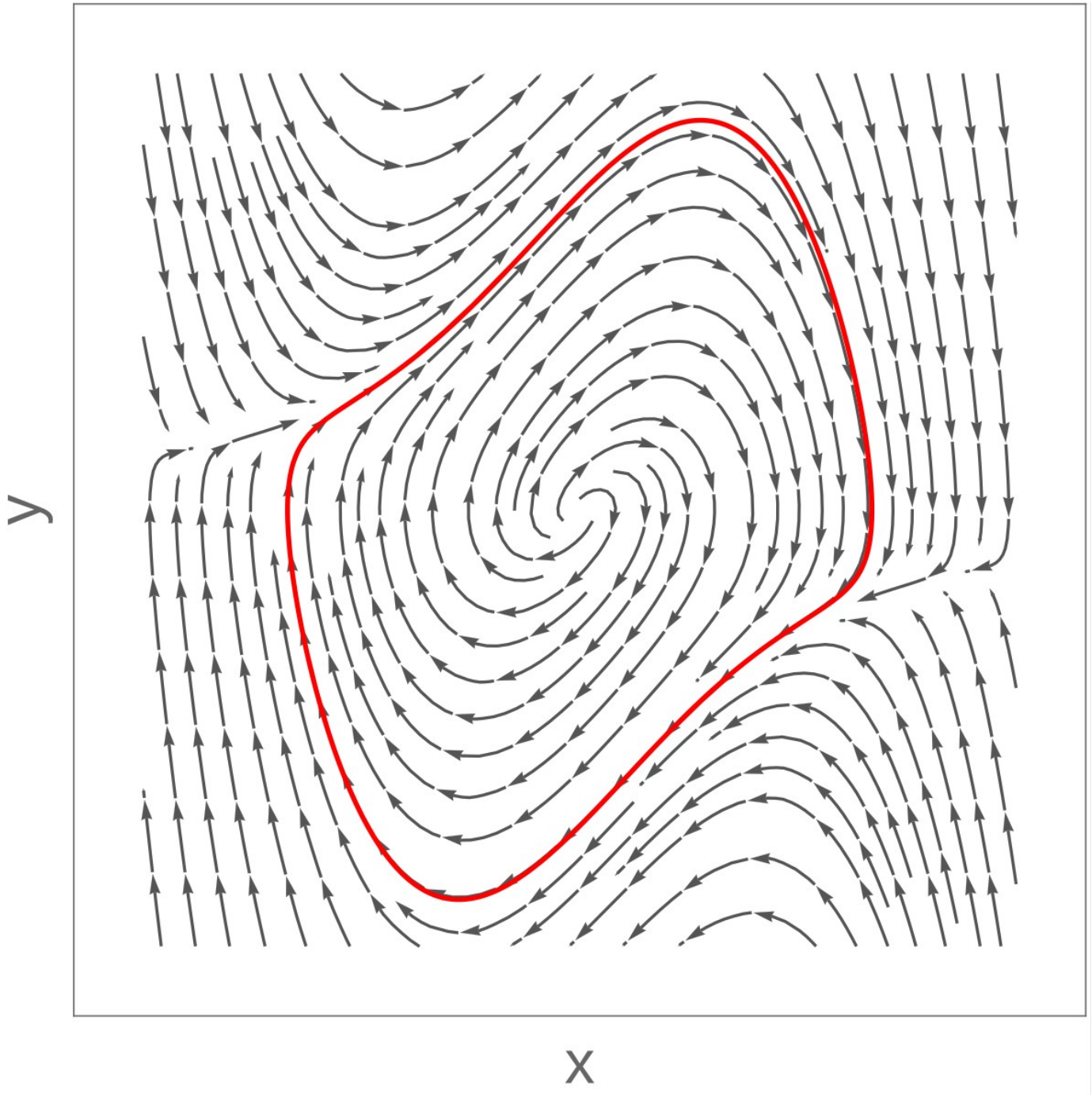


stable node

attracting



Limit cycle



https://arxiv.org/abs/1710.11029
SGD limit cycles for deep nets?
(I guess follow-up work said no)

Chaos –

strange attractor

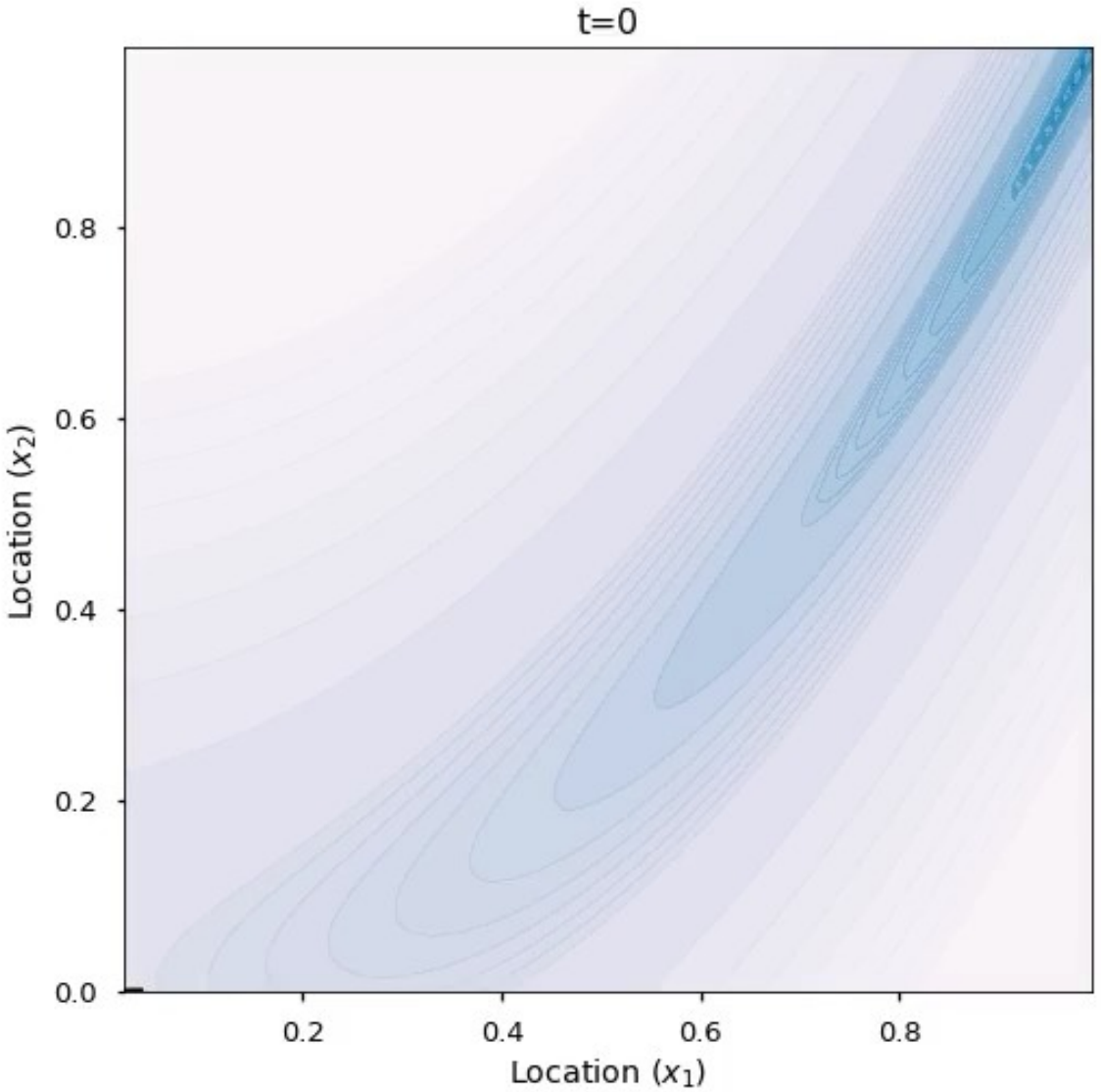
Back to physics

- Newton:

Two dots!

One dot!

Classic “one dot” slow down (gradient descent)



The difference has to do with *momentum*

“Momentum” in ML is really useful for good results with SGD

CIFAR-100 hyper-parameter tuning experiment: https://github.com/meliketoy/wide-resnet.pytorch

epoch	learning rate	weight decay	Optimizer	Momentum	Nesterov
0 ~ 60	0.1	0.0005	Momentum	0.9	true
61 ~ 120	0.02	0.0005	Momentum	0.9	true
121 ~ 160	0.004	0.0005	Momentum	0.9	true
161 ~ 200	0.0008	0.0005	Momentum	0.9	true

network	dropout	preprocess	GPU:0	GPU:1	per epoch	Top1 acc(%)	Top5 acc(%)
wide-resnet 28x20	0.3	meanstd	8.13G	6.93G	4 min 05 sec	**82.45**	**96.11**

Momentum - Newton to Hamilton

•

Beauty of Hamilton's formulation

•

"position", "momentum" or "velocity"

$$H(x,v) = E(x) + K(v)$$

Hamiltonian — Potential energy — Kinetic energy

Hamiltonian explicitly connects conservation laws and dynamics

"position" "momentum" or "velocity"

$$H(x, v) = E(x) + K(v)$$

Hamiltonian dynamics

Hamiltonian

Potential energy

Kinetic energy

How do we know the energy is conserved for these dynamics?

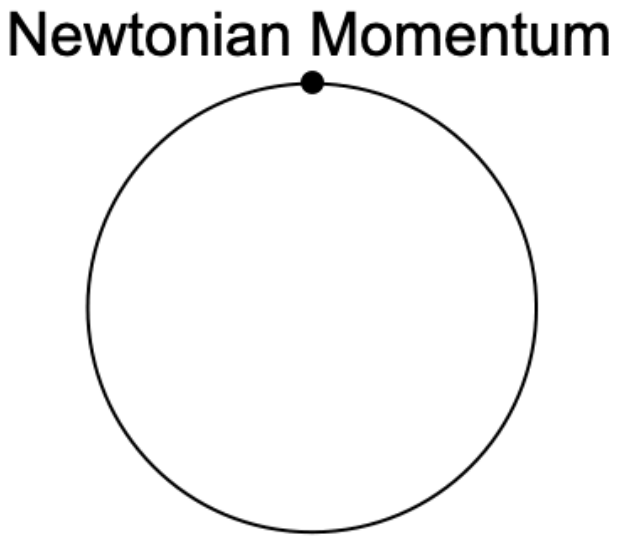
$$\dot{H} = dH(\mathbf{x}, \mathbf{v})/dt = \frac{\partial H}{\partial \mathbf{x}} \cdot \frac{d\mathbf{x}}{dt} + \frac{\partial H}{\partial \mathbf{v}} \cdot \frac{d\mathbf{v}}{dt}$$

$$\dot{H} = \frac{\partial H}{\partial \mathbf{x}} \cdot \dot{\mathbf{x}} + \frac{\partial H}{\partial \mathbf{v}} \cdot \dot{\mathbf{v}}$$

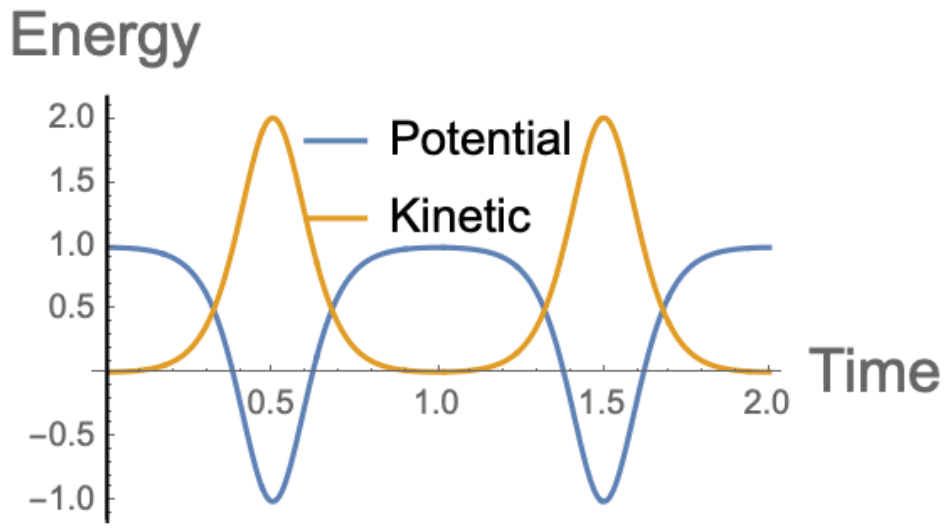
$$= \frac{\partial H}{\partial \mathbf{x}} \cdot \frac{\partial H}{\partial \mathbf{v}} + \frac{\partial H}{\partial \mathbf{v}} \cdot \left(-\frac{\partial H}{\partial \mathbf{x}}\right)$$

$$\dot{H} = 0$$

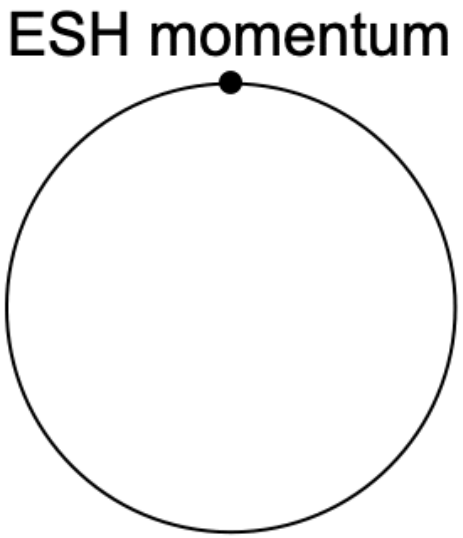
Roller coaster momentum and conservation

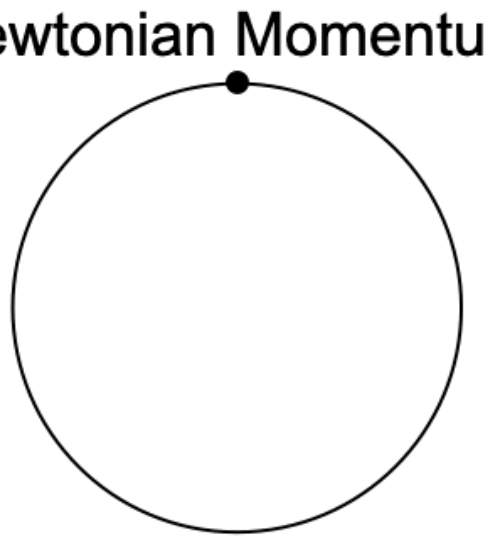


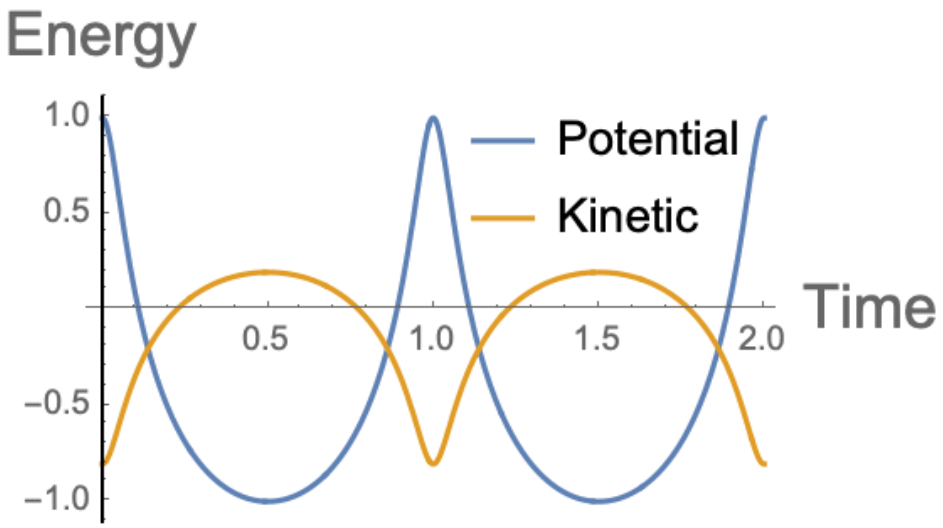
This momentum isn’t good for optimization at all.

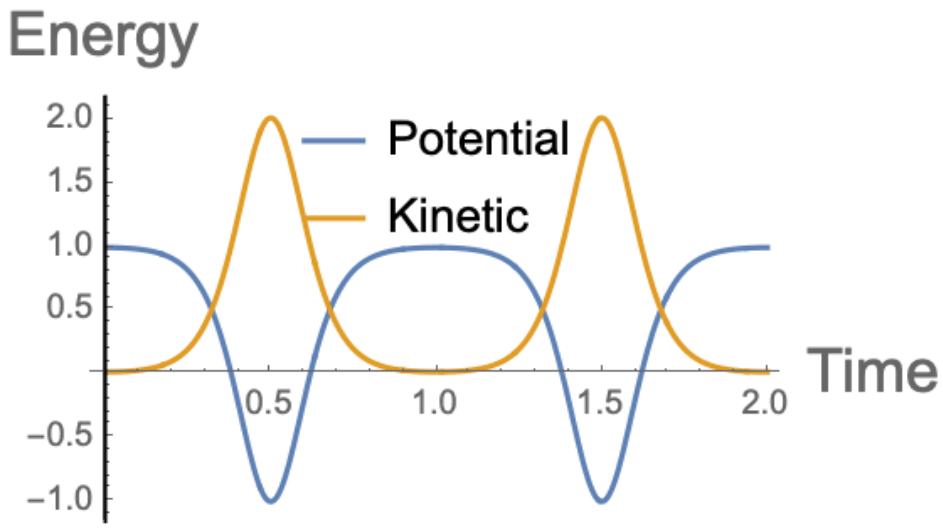


(BTW: the sampling paper I mentioned uses a non-Newtonian momentum
https://arxiv.org/abs/2111.02434
I originally made this figure for that.)









“Momentum” in ML is momentum + friction

Classical dynamics
with momentum

Classical dynamics
with momentum and friction

•

“Momentum” in ML is momentum + friction

Classical dynamics
with momentum

Classical dynamics
with momentum and friction

•

On the importance of initialization and momentum in deep learning

Ilya Sutskever[1] ILYASU@GOOGLE.COM

2. Momentum and Nesterov's Accelerated Gradient

The momentum method (Polyak, 1964), which we refer to as classical momentum (CM), is a technique for accelerating gradient descent that accumulates a velocity vector in directions of persistent reduction in the objective across iterations. Given an objective function $f(\theta)$ to be minimized, classical momentum is given by:

$$v_{t+1} = \mu v_t - \varepsilon \nabla f(\theta_t) \quad (1)$$

$$\theta_{t+1} = \theta_t + v_{t+1} \quad (2)$$

On the importance of initialization and momentum in deep learning

Ilya Sutskever[1] ILYASU@GOOGLE.COM

$$v_{t+1} = \mu v_t - \varepsilon \nabla f(\theta_t) \quad (1)$$
$$\theta_{t+1} = \theta_t + v_{t+1} \quad (2)$$

- Classical momentum

cally, as shown in the appendix, the NAG update may be rewritten as:

$$v_{t+1} = \mu v_t - \varepsilon \nabla f(\theta_t + \mu v_t) \quad (3)$$
$$\theta_{t+1} = \theta_t + v_{t+1} \quad (4)$$

- Nesterov accelerated gradient (NAG) has optimal convergence for convex optimization

Helpful (?) illustration from Sutskever et al.

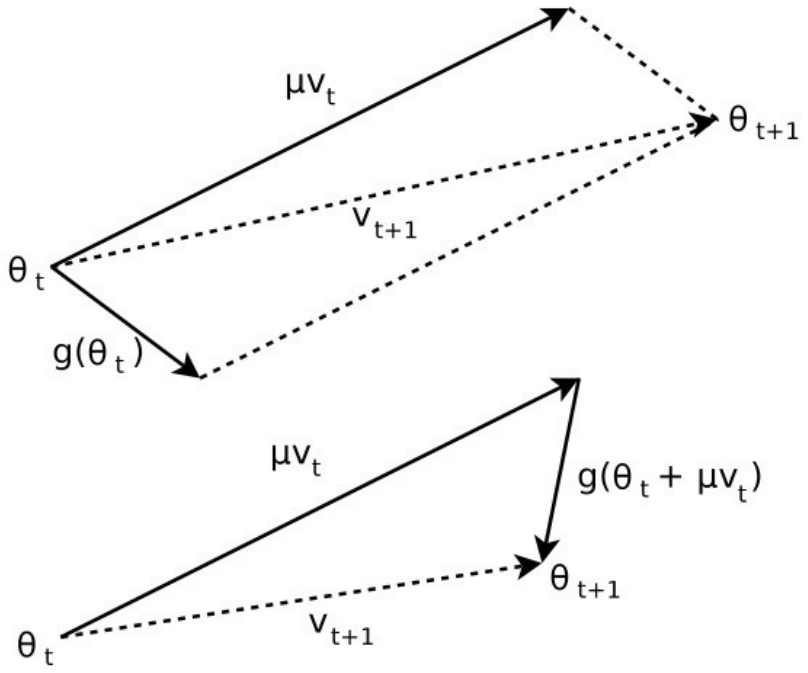


Figure 1. **(Top)** Classical Momentum **(Bottom)** Nesterov Accelerated Gradient

More helpful (to me)

$$\ddot{X} + \frac{3}{t}\dot{X} + \nabla f(X) = 0$$

Journal of Machine Learning Research 17 (2016) 1-43 Submitted 3/15; Revised 10/15; Published 9/16

A Differential Equation for Modeling Nesterov's Accelerated Gradient Method: Theory and Insights

Weijie Su SUW@WHARTON.UPENN.EDU
Department of Statistics
University of Pennsylvania
Philadelphia, PA 19104, USA

Stephen Boyd BOYD@STANFORD.EDU
Department of Electrical Engineering
Stanford University
Stanford, CA 94305, USA

Emmanuel J. Candès CANDES@STANFORD.EDU
Departments of Statistics and Mathematics
Stanford University
Stanford, CA 94305, USA

Editor: Yoram Singer

Abstract

We derive a second-order ordinary differential equation (ODE) which is the limit of Nesterov's accelerated gradient method. This ODE exhibits approximate equivalence to Nesterov's scheme and thus can serve as a tool for analysis. We show that the continuous time ODE allows for a better understanding of Nesterov's scheme. As a byproduct, we obtain a family of schemes with similar convergence rates. The ODE interpretation also suggests restarting Nesterov's scheme leading to an algorithm, which can be rigorously proven to converge at a linear rate whenever the objective is strongly convex.

Keywords: Nesterov's accelerated scheme, convex optimization, first-order methods, differential equation, restarting

Warning – parameters of momentum implementation don’t match across frameworks

- https://pytorch.org/docs/stable/generated/torch.optim.SGD.html

NOTE

The implementation of SGD with Momentum/Nesterov subtly differs from Sutskever et. al. and implementations in some other frameworks.

Considering the specific case of Momentum, the update can be written as

$$v_{t+1} = \mu * v_t + g_{t+1},$$
$$p_{t+1} = p_t - \text{lr} * v_{t+1},$$

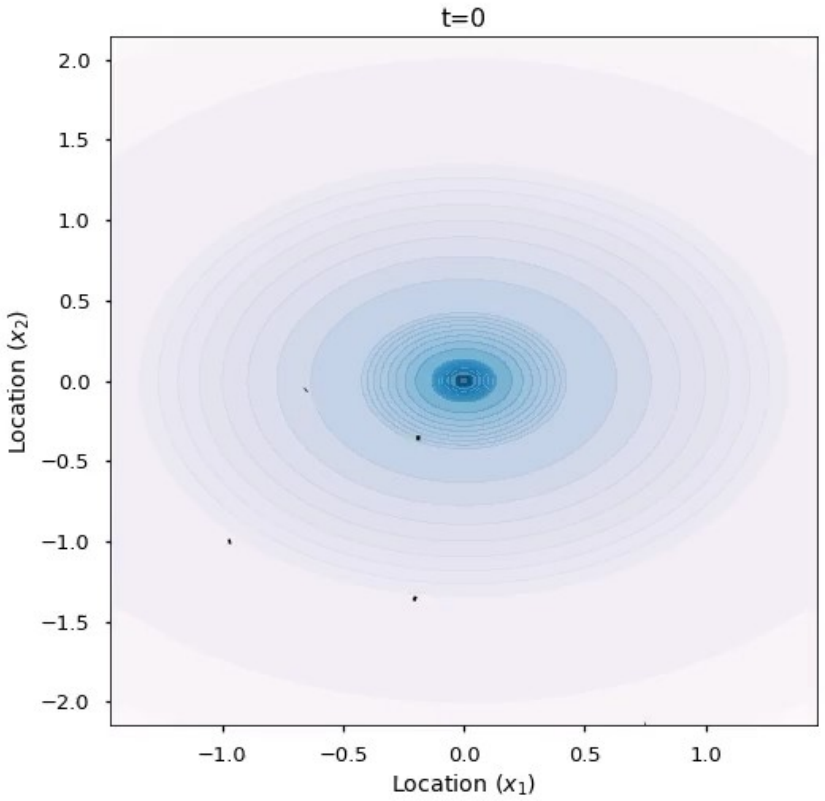
where p, g, v and μ denote the parameters, gradient, velocity, and momentum respectively.

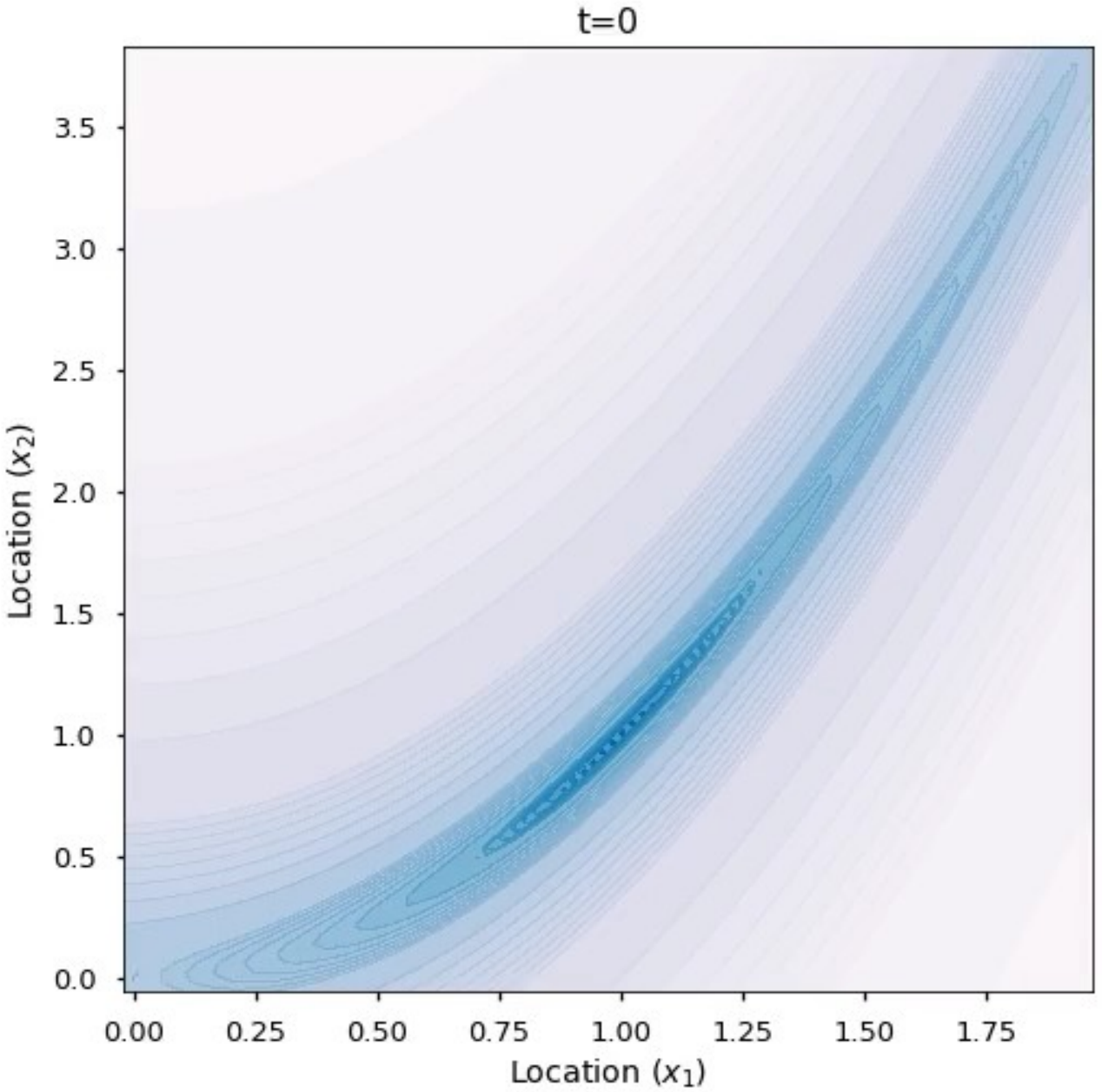
This is in contrast to Sutskever et. al. and other frameworks which employ an update of the form

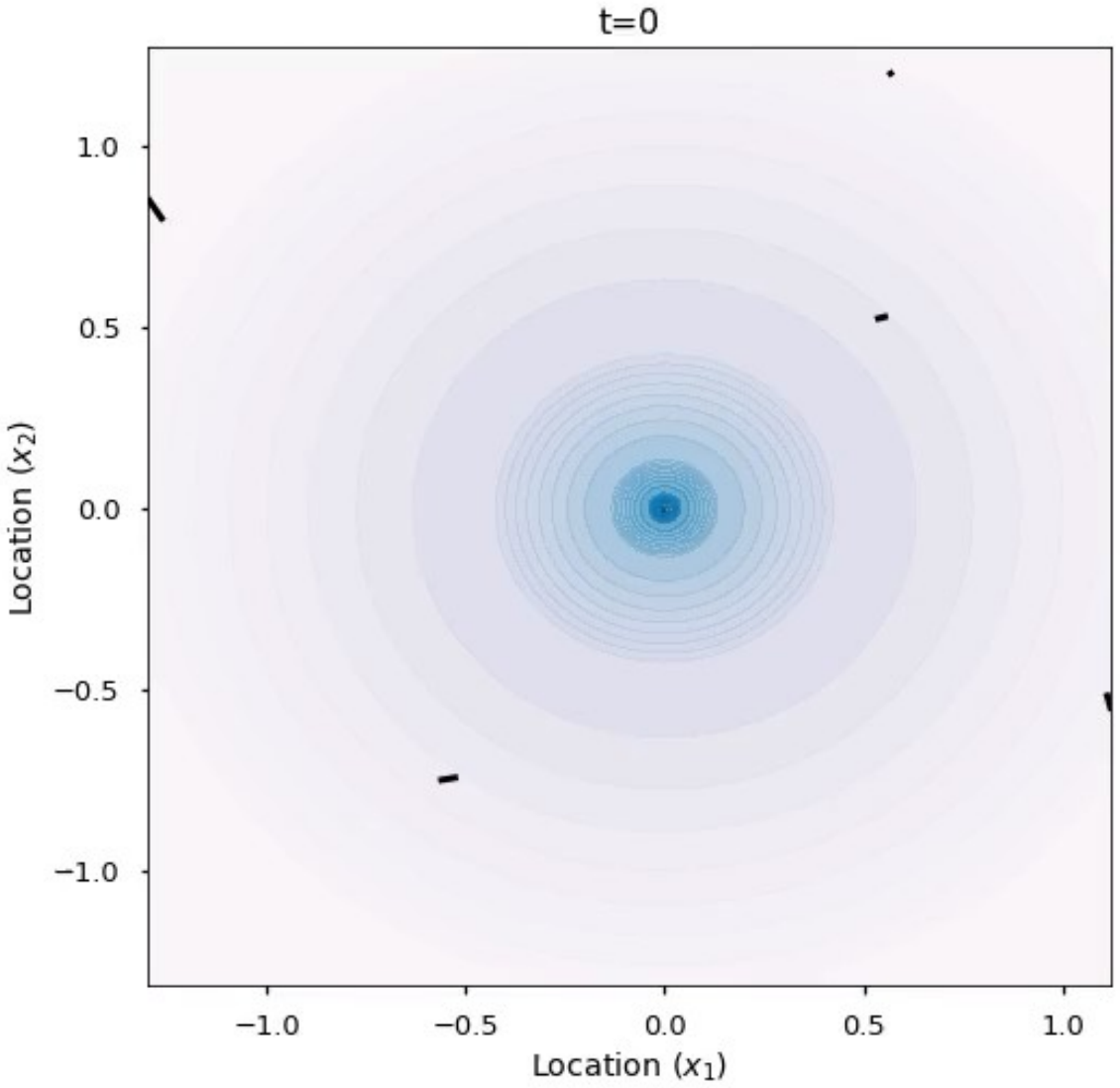
$$v_{t+1} = \mu * v_t + \text{lr} * g_{t+1},$$
$$p_{t+1} = p_t - v_{t+1}.$$

The Nesterov version is analogously modified.

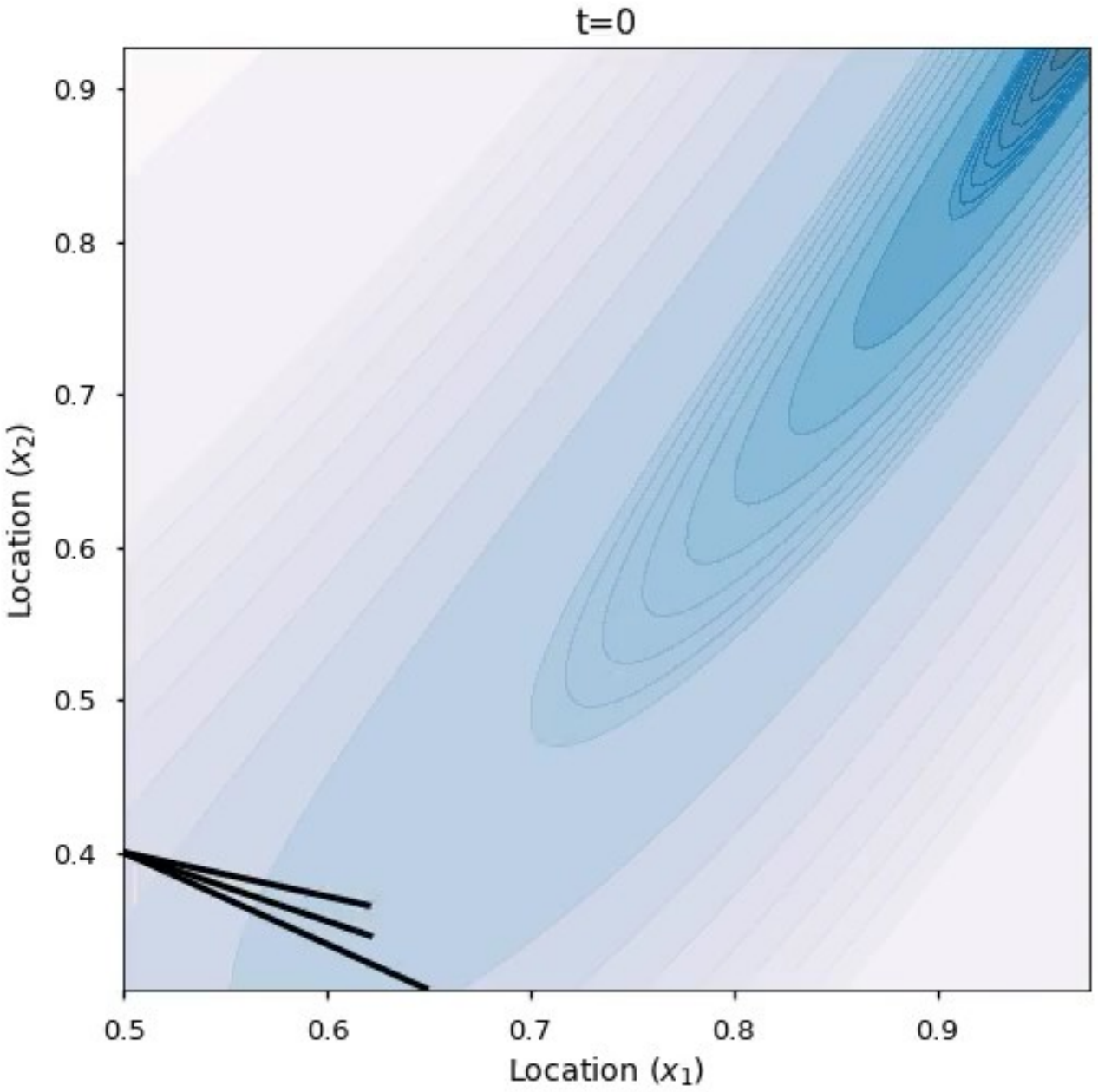
Hamiltonian dynamics (momentum but no friction)







- Hamiltonian dynamics with friction
- Classic (static) friction slows things down too much!
- (Hence default is “nesterov” momentum)



- Using large step size is implicitly like adding acceleration, but can lead to numerical errors
- Noether dynamics paper, NeurIPS 2021

Wednesday: SGD and why it really works

•

Sum of iid random variables

For large enough batch size, central limit theorem describes distribution

Wednesday: SGD and why it really works

•